

Calibration of Production Transient Alert Brokers: Preparing Decision Infrastructure for LSST

Can broker confidence scores be trusted for automated follow-up decisions?

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QR code

Code, tables & figures
branch [alerce-selective-deferral](#)

1 Motivation

“When a broker reports 80% confidence, is it actually correct 80% of the time?”

Millions of alerts → Broker class probabilities → Gates & rankings → Follow-up decisions

→ Limited telescope resources

- Rubin/LSST alert volumes cannot be inspected manually; broker probabilities feed rankings, automated filters, human review and follow-up requests.
- A classifier may rank the correct class first while reporting unreliable confidence values.
- Miscalibration either rejects useful candidates or auto-accepts unreliable labels.

Underconfidence

Correct classifications score too low → useful objects fail fixed probability gates.

Overconfidence

Incorrect classifications score too high → unreliable labels trusted automatically.

A functioning classifier is not automatically functioning decision infrastructure.

2 Dataset & ground truth

Spectroscopically confirmed ZTF Bright Transient Survey objects, cross-matched with the latest available 15-class ALeRCE light-curve probability vectors.

1,606

valid cross-matched objects

1,576

true class exists in deployed ALeRCE taxonomy

30

TDEs — taxonomy-gap stress test

SN Ia 757

SN II 560

SN Ibc 204

SLSN 55

TDE 30 — excluded from in-taxonomy analysis

- BTS spectroscopy = ground truth; ALeRCE = 15-class probability vector.
- SN Ia, SN II, SN Ibc, SLSN exist in the deployed taxonomy; **TDE does not** — the 30 TDEs test whether the pipeline flags an *unrepresentable* object as unreliable.

Per-class top-1 accuracy

SN Ia 83.1%

SN II 49.5%

SN Ibc 82.8%

SLSN 78.2%

Overall accuracy 70.9%, but performance differs strongly by class — SN II is the main classification bottleneck.

6 What would this change in a real pipeline?

ALeRCE probabilities → Calibration layer → Reliability checker → Ranked decision queue

High reliability

- Auto-accept broker label
- Routine photometric monitoring
- No unnecessary manual review

Low reliability

- Wait for more detections / second broker
- Human review, host-galaxy context
- Spectroscopy only when scientifically valuable

For the same review budget, the checker concentrates far more errors into the deferred queue than raw confidence.

Historical alert-stream replay is required to establish real-time telescope-level benefit. Next: reconstruct probabilities after 2, 5, 10 detections · fixed nightly budget · errors found per 100 reviewed · rare-transient recovery · later observing season.

3 Calibration-aware selective-classification pipeline

- Compare labels.** ALeRCE top-ranked class vs independent spectroscopic label.
- Define correctness.** Correct = 1 if top-1 class matches spectroscopy, else 0.
- Audit calibration.** Predictions near confidence 0.7 should be correct ≈ 70% of the time.

$$ECE = \sum_b (n_b/N) \cdot |acc(b) - conf(b)|$$

- Recalibrate.** Held-out temperature scaling repairs the probability scale — *it never changes the top-ranked class*.

$$\hat{p} = \text{softmax}(\log p / T)$$

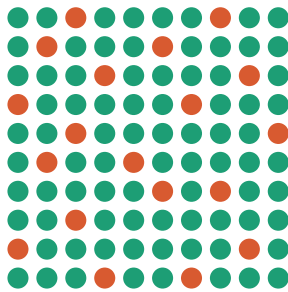
- Uncertainty features.** Top-1, top-2, margin, normalised entropy, transient-branch mass, conditional transient confidence, non-transient competition.
- Train the checker.** Logistic regression estimating **P(top-1 is correct)**: **learned_shape** (shape features) and **learned_full** (all 15 calibrated probabilities + shape).
- Accept or defer.** High score → auto-accept label. Low score → defer for more detections, another broker, human review, or justified spectroscopy.

$$\text{Chow's rule: } \text{defer} \Leftrightarrow \hat{P}(\text{error}) > c_{\text{defer}} \text{ (Chow 1970)}$$

“Defer” does not automatically mean spectroscopy.

Evaluation: five-fold out-of-fold (train 80% / test unseen 20%) · paired stratified bootstrap · ROC-AUC · AURC & risk-coverage · fixed-threshold operating points · TDE stress test. *Post-hoc selective classification — not a jointly trained learning-to-defer system, not a deployed real-time LSST pipeline.*

4 Calibration and error detection are different



100 predictions at confidence 0.8 → 80 correct, 20 incorrect.

A group may be perfectly calibrated even when the score cannot identify *which individual predictions* are wrong.

Same top confidence, different structure

Object A — clean

SN Ia 0.70 · SN II 0.10 · SN Ibc 0.08 · others 0.12

Object B — contested

SN Ia 0.70 · **SN II 0.28** · others 0.02

Identical top confidence, but Object B has a strong competing class. The checker uses the complete probability pattern.

ALeRCE: “What class is this?”

Checker: “Should this classification be trusted automatically?”

5 Results

AGGREGATE CALIBRATION

0.259 → 0.015

ECE — aggregate reliability strongly improved after held-out temperature scaling.

CHECKER RANKING QUALITY

0.836

ROC-AUC (correct vs incorrect). Class-rate control 0.752 · raw confidence 0.689.

CHECKER VS CLASS CONTROL

+0.083

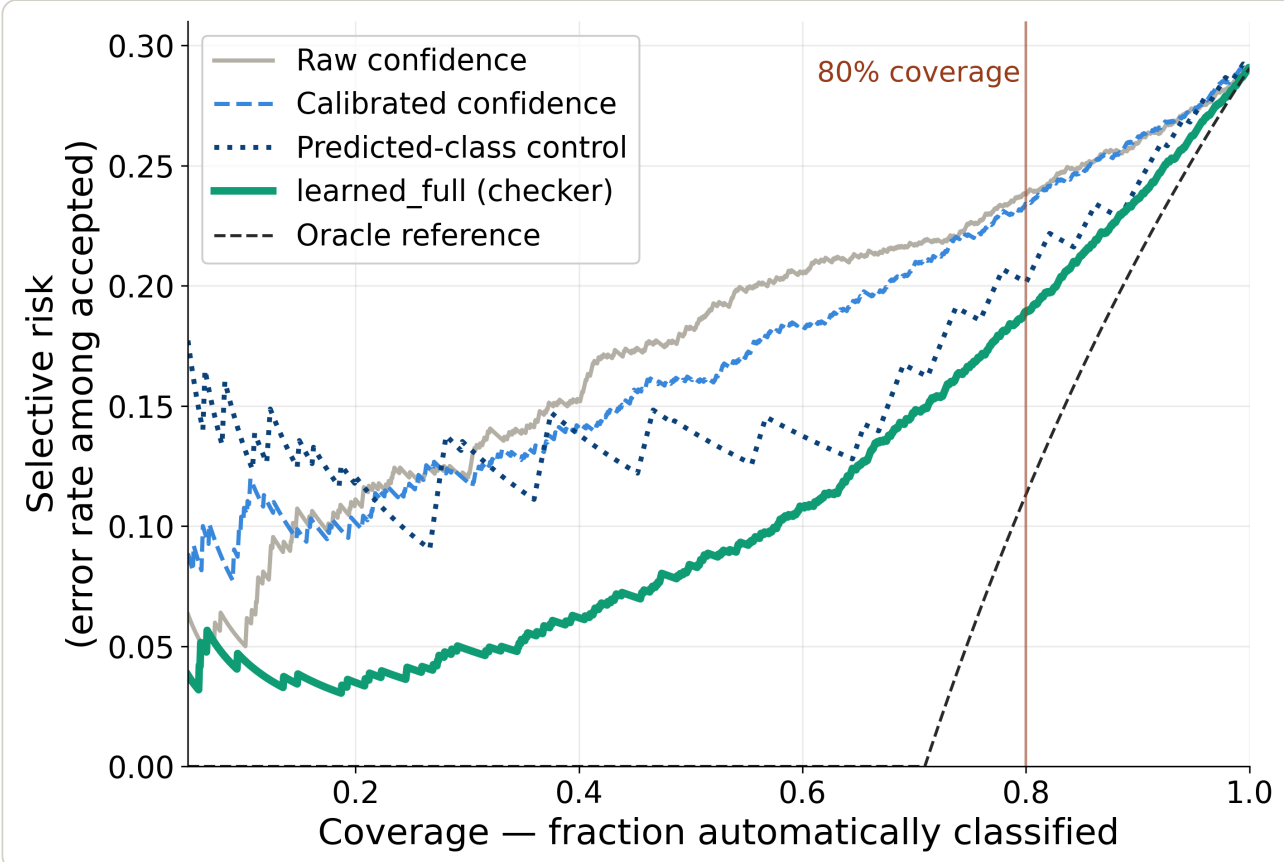
Δ ROC-AUC, paired bootstrap 95% CI [0.057, 0.107] — object-level signal beyond class identity.

AT 80% AUTO-ACCEPT COVERAGE

81.0% · 47.8%

Accepted accuracy · errors captured in the deferred 20%.

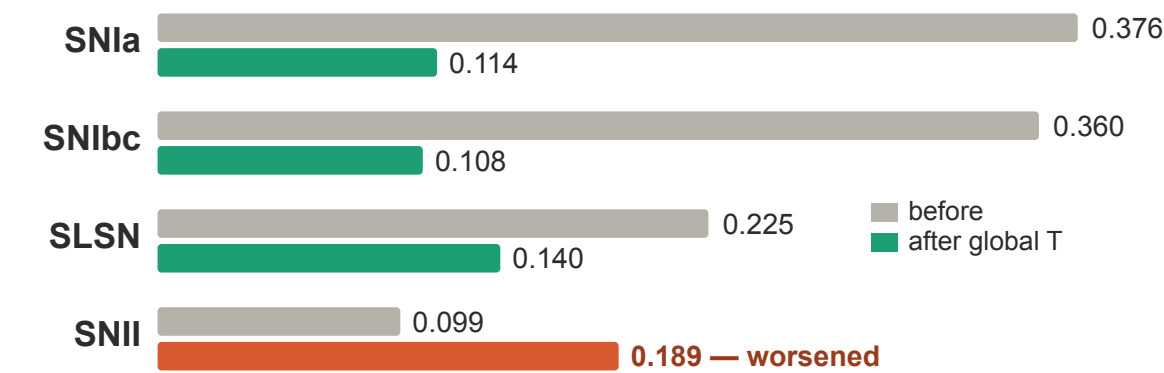
Risk-coverage: the checker gives the lowest risk



Policy @ 80% coverage	Accepted acc.	Errors captured
Raw confidence	76.2%	34.5%
Predicted-class control	79.9%	44.5%
learned_full (checker)	81.0%	47.8%

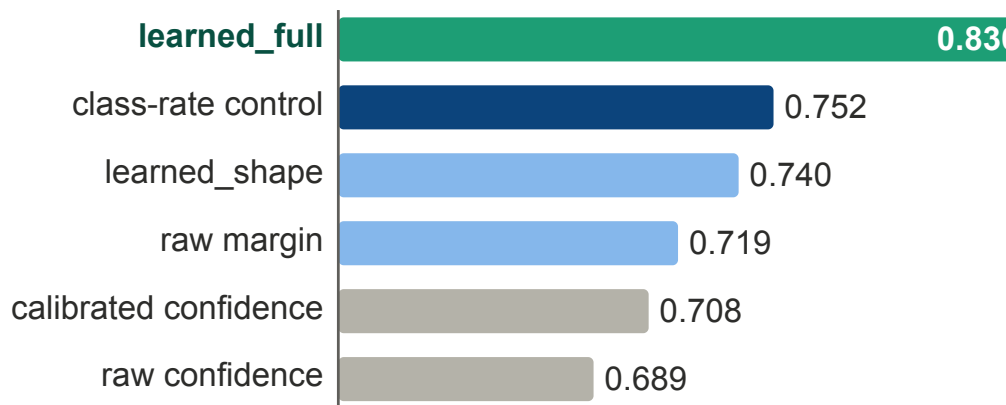
Lowest selective risk across the operational coverage range — AURC 0.115 (checker) vs 0.173 (raw confidence).

Class-wise ECE: one global correction is not enough



SN Ia/SN Ibc: accurate but strongly underconfident — repaired. SN II: calibration **worsened** by the global correction. SLSN: only 55 objects. Class-dependent miscalibration remains.

Correctness ranking (ROC-AUC)



The shape-only model does not clearly beat the class-rate control — the full 15-class probability vector supplies the additional signal.

Calibration changes threshold meaning

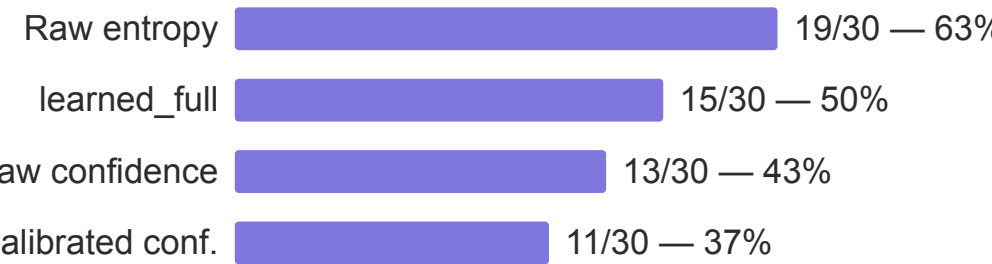
Raw $p \geq 0.8$ 9 accepted · 0.6% cov. · 100% acc. (only nine objects)

Calibrated $p \geq 0.8$ 395 accepted · 25.1% cov. · 88.4% accuracy

≈ 44-fold larger acceptance set at the same numerical gate

Calibration mainly repairs the numerical probability scale — it does not create new correct classifications or substantially reorder objects.

Missing classes need a separate novelty channel



TDEs deferred at the 80%-coverage cutoff. **None of the confidence-based policies reliably detected the taxonomy gap** (n = 30; intervals overlap).

The policy that works best for ordinary in-taxonomy errors is not necessarily the best novelty detector.

Known-class channel: calibrated probability + checker

Novelty channel: explicit OOD / anomaly detector

Simple cost model (deferral cost 0.20, unit error cost): learned_full 0.142/object · class control 0.161 · calibrated conf. 0.179 · raw 0.199 · defer-all 0.200. Generic costs — not a class-weighted telescope utility model.

Main references

- Sánchez-Sáez et al. 2021, *The ALeRCE light curve classifier*, AJ.
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El-Yaniv & Wiener 2010, *Foundations of noise-free selective classification*, JMLR.
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7 Conclusions

- ALeRCE frequently ranks the correct class first, but its confidence values are systematically too low.
- Calibration repairs the interpretation of numerical probability gates without changing the top-ranked class.
- A post-hoc checker using the complete 15-class probability vector identifies unreliable predictions better than raw confidence or predicted-class identity alone.
- Confidence-based selective classification cannot replace explicit novelty detection when the true class is absent from the taxonomy.

Calibrated probabilities decide how much to trust a prediction. A reliability checker decides which predictions to trust. A novelty detector is still needed for classes the broker cannot represent.

A functioning alert classifier is not automatically functioning decision infrastructure.